

# Class 13: Transcriptomics and the analysis of RNA-Seq data

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## Background

The data for this hands-on session comes from a published RNA-seq experiment where airway smooth muscle cells were treated with dexamethasone, a synthetic glucocorticoid steroid with anti-inflammatory effects (Himes et al. 2014).

## Bioconductor setup

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,  
match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,  
Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,  
table, tapply, union, unique, unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: GenomeInfoDb

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,  
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,  
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,  
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,  
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,  
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,  
colWeightedMeans, colWeightedMedians, colWeightedSds,  
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,  
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,  
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,  
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,  
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,  
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,  
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,  
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with  
'browseVignettes()'. To cite Bioconductor, see  
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
# Read in the data
counts <- read.csv("airway_scaledcounts.csv", row.names=1)
metadata <- read.csv("airway_metadata.csv")

head(counts)
```

|                  | SRR1039508 | SRR1039509 | SRR1039512 | SRR1039513 | SRR1039516 |
|------------------|------------|------------|------------|------------|------------|
| ENSG000000000003 | 723        | 486        | 904        | 445        | 1170       |
| ENSG000000000005 | 0          | 0          | 0          | 0          | 0          |
| ENSG000000000419 | 467        | 523        | 616        | 371        | 582        |
| ENSG000000000457 | 347        | 258        | 364        | 237        | 318        |
| ENSG000000000460 | 96         | 81         | 73         | 66         | 118        |
| ENSG000000000938 | 0          | 0          | 1          | 0          | 2          |

|                  | SRR1039517 | SRR1039520 | SRR1039521 |
|------------------|------------|------------|------------|
| ENSG000000000003 | 1097       | 806        | 604        |
| ENSG000000000005 | 0          | 0          | 0          |
| ENSG000000000419 | 781        | 417        | 509        |
| ENSG000000000457 | 447        | 330        | 324        |
| ENSG000000000460 | 94         | 102        | 74         |
| ENSG000000000938 | 0          | 0          | 0          |

```
head(metadata)
```

|   | id         | dex     | celltype | geo_id     |
|---|------------|---------|----------|------------|
| 1 | SRR1039508 | control | N61311   | GSM1275862 |
| 2 | SRR1039509 | treated | N61311   | GSM1275863 |
| 3 | SRR1039512 | control | N052611  | GSM1275866 |
| 4 | SRR1039513 | treated | N052611  | GSM1275867 |
| 5 | SRR1039516 | control | N080611  | GSM1275870 |
| 6 | SRR1039517 | treated | N080611  | GSM1275871 |

Q1. How many genes are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

Q2. How many 'control' cell lines do we have?

```
table(metadata$dex)
```

```
control treated
      4      4
```

## Toy differential gene expression

```
# Get control samples
control <- metadata[metadata[, "dex"]=="control",]
control.counts <- counts[,control$id]
control.mean <- rowSums(control.counts)/4
head(control.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460
      900.75          0.00          520.50          339.75          97.25
ENSG000000000938
      0.75
```

```
# Get treated samples
treated <- metadata[metadata[, "dex"]=="treated",]
treated.counts <- counts[,treated$id]
treated.mean <- rowSums(treated.counts)/4
head(treated.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460
      658.00          0.00          546.00          316.50          78.75
ENSG000000000938
      0.00
```

Q3. How would you make the above code in either approach more robust?  
Is there a function that could help here?

Replace /4 with /ncol

Q4. Follow the same procedure for the treated samples (i.e. calculate the mean per gene across drug treated samples and assign to a labeled vector called treated.mean)

```
treated <- metadata[metadata[, "dex"]=="treated",]
treated.counts <- counts[,treated$id]
treated.mean <- rowSums(treated.counts)/ncol(treated.counts)
head(treated.mean)
```

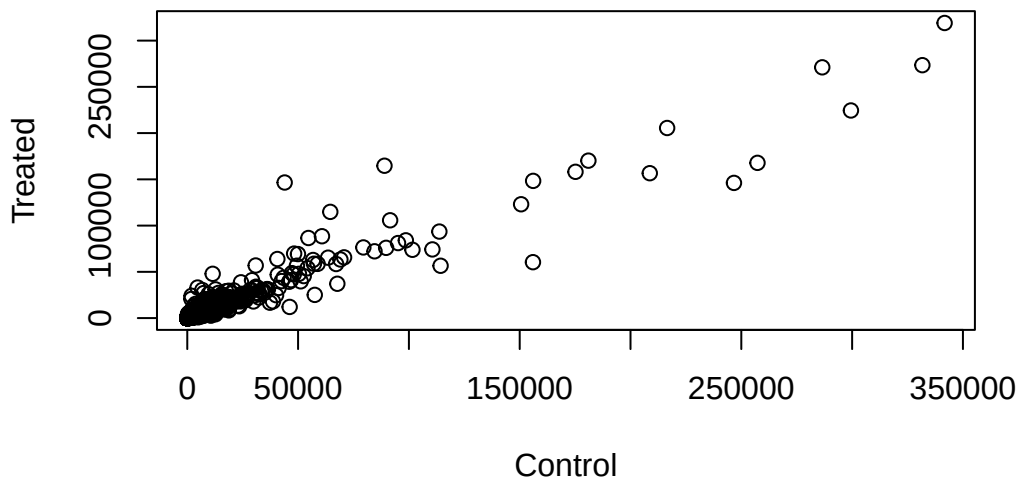
```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460
        658.00           0.00          546.00          316.50          78.75
ENSG000000000938
        0.00
```

```
meancounts <- data.frame(control.mean, treated.mean)
colSums(meancounts)
```

```
control.mean treated.mean
    23005324     22196524
```

Q5 (a). Create a scatter plot showing the mean of the treated samples against the mean of the control samples. Your plot should look something like the following.

```
plot(meancounts[,1], meancounts[,2],
     xlab="Control", ylab="Treated")
```



Q5 (b). You could also use the ggplot2 package to make this figure producing the plot below. What `geom_?()` function would you use for this plot?

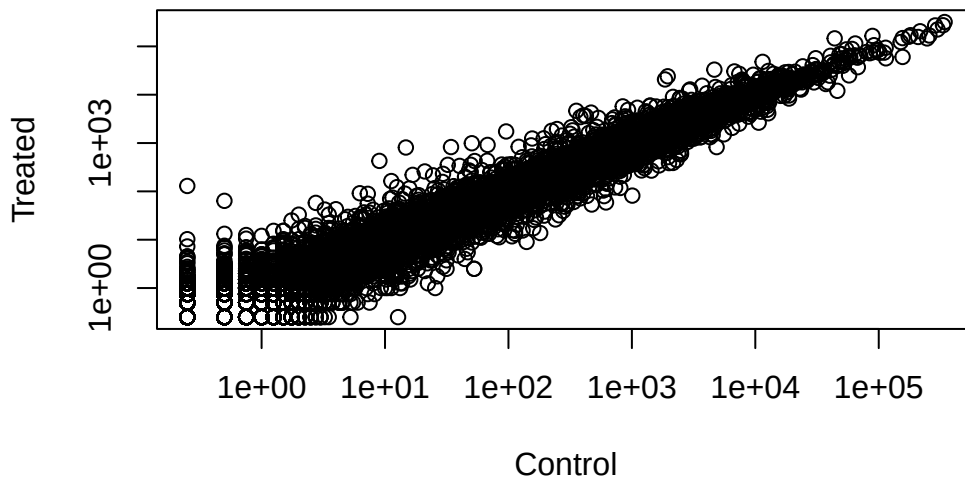
Use `geom_point()`

Q6. Try plotting both axes on a log scale. What is the argument to `plot()` that allows you to do this?

```
plot(meancounts[,1], meancounts[,2],  
     xlab="Control", ylab="Treated",  
     log="xy")
```

Warning in `xy.coords(x, y, xlabel, ylabel, log)`: 15032 x values  $\leq 0$  omitted from logarithmic plot

Warning in `xy.coords(x, y, xlabel, ylabel, log)`: 15281 y values  $\leq 0$  omitted from logarithmic plot



The `log="xy"` argument sets both axes to log scale. This is useful because most genes cluster near zero and the log scale spreads them out so we can see the differences better.

```
# Add log2 fold change
meancounts$log2fc <- log2(meancounts[, "treated.mean"]/meancounts[, "control.mean"])
head(meancounts)
```

|                  | control.mean | treated.mean | log2fc      |
|------------------|--------------|--------------|-------------|
| ENSG000000000003 | 900.75       | 658.00       | -0.45303916 |
| ENSG000000000005 | 0.00         | 0.00         | NaN         |
| ENSG000000000419 | 520.50       | 546.00       | 0.06900279  |
| ENSG000000000457 | 339.75       | 316.50       | -0.10226805 |
| ENSG000000000460 | 97.25        | 78.75        | -0.30441833 |
| ENSG000000000938 | 0.75         | 0.00         | -Inf        |

```
# Remove zero count genes
zero.vals <- which(meancounts[,1:2]==0, arr.ind=TRUE)
to.rm <- unique(zero.vals[,1])
mycounts <- meancounts[-to.rm,]
```

Q7. What is the purpose of the `arr.ind` argument in the `which()` function call above? Why would we then take the first column of the output and need to call the `unique()` function?

The `arr.ind=TRUE` argument makes `which()` return both the row and column positions of zero values, not just a single index. Take the first column (rows) and use `unique()` to avoid counting a gene twice if it has zeros in both samples.

```
up.ind <- mycounts$log2fc > 2
sum(up.ind)
```

```
[1] 250
```

Q8. Using the `up.ind` vector above can you determine how many up regulated genes we have at the greater than 2 fc level?

250 genes are upregulated ( $\log_2\text{fc} > 2$ )

```
down.ind <- mycounts$log2fc < (-2)
sum(down.ind)
```

```
[1] 367
```

Q9. Using the `down.ind` vector above can you determine how many down regulated genes we have at the greater than 2 fc level?



367 genes are downregulated ( $\log_2\text{fc} < -2$ )

Q10. Do you trust these results? Why or why not?

No because we only looked at fold change and not statistical testing. A gene can have a large fold change but still not be statistically significant. We need DESeq2 to properly assess this.

## Setting up for DESeq

```
dds <- DESeqDataSetFromMatrix(countData=counts,  
                               colData=metadata,  
                               design=~dex)
```

converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds
```

```
class: DESeqDataSet  
dim: 38694 8  
metadata(1): version  
assays(1): counts  
rownames(38694): ENSG000000000003 ENSG000000000005 ... ENSG00000283120  
               ENSG00000283123  
rowData names(0):  
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521  
colData names(4): id dex celltype geo_id
```

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds)
res
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 38694 rows and 6 columns

|                  | baseMean  | log2FoldChange | lfcSE     | stat      | pvalue    |
|------------------|-----------|----------------|-----------|-----------|-----------|
|                  | <numeric> | <numeric>      | <numeric> | <numeric> | <numeric> |
| ENSG000000000003 | 747.1942  | -0.3507030     | 0.168246  | -2.084470 | 0.0371175 |
| ENSG000000000005 | 0.0000    | NA             | NA        | NA        | NA        |
| ENSG000000000419 | 520.1342  | 0.2061078      | 0.101059  | 2.039475  | 0.0414026 |
| ENSG000000000457 | 322.6648  | 0.0245269      | 0.145145  | 0.168982  | 0.8658106 |
| ENSG000000000460 | 87.6826   | -0.1471420     | 0.257007  | -0.572521 | 0.5669691 |
| ...              | ...       | ...            | ...       | ...       | ...       |
| ENSG00000283115  | 0.000000  | NA             | NA        | NA        | NA        |
| ENSG00000283116  | 0.000000  | NA             | NA        | NA        | NA        |
| ENSG00000283119  | 0.000000  | NA             | NA        | NA        | NA        |
| ENSG00000283120  | 0.974916  | -0.668258      | 1.69456   | -0.394354 | 0.693319  |
| ENSG00000283123  | 0.000000  | NA             | NA        | NA        | NA        |
|                  | padj      |                |           |           |           |
|                  | <numeric> |                |           |           |           |
| ENSG000000000003 | 0.163035  |                |           |           |           |
| ENSG000000000005 | NA        |                |           |           |           |
| ENSG000000000419 | 0.176032  |                |           |           |           |
| ENSG000000000457 | 0.961694  |                |           |           |           |
| ENSG000000000460 | 0.815849  |                |           |           |           |
| ...              | ...       |                |           |           |           |
| ENSG00000283115  | NA        |                |           |           |           |
| ENSG00000283116  | NA        |                |           |           |           |
| ENSG00000283119  | NA        |                |           |           |           |
| ENSG00000283120  | NA        |                |           |           |           |
| ENSG00000283123  | NA        |                |           |           |           |

```
summary(res)
```

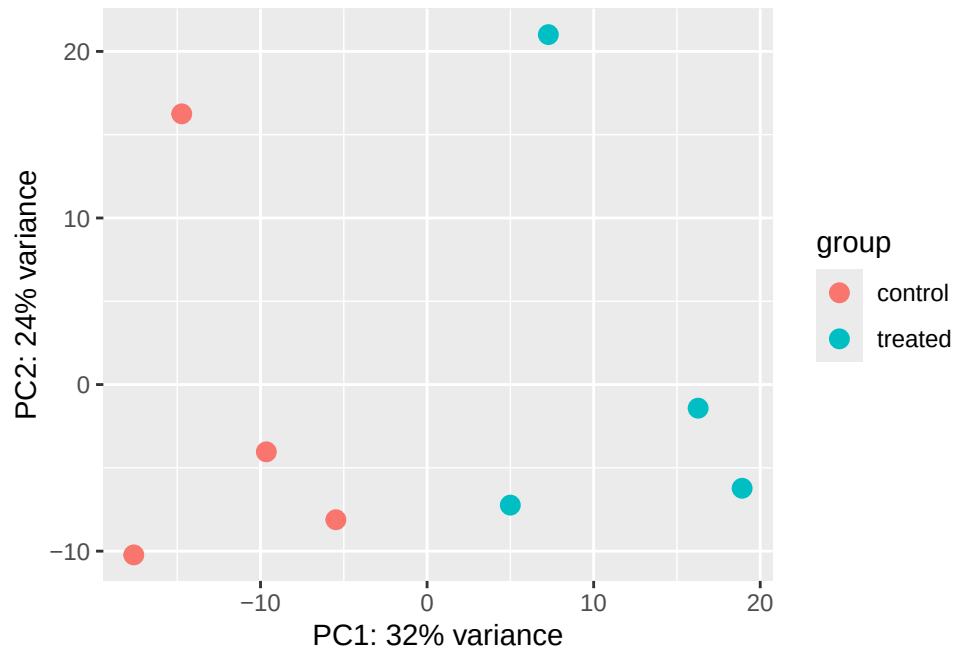
```
out of 25258 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 1563, 6.2%
LFC < 0 (down)    : 1188, 4.7%
outliers [1]      : 142, 0.56%
low counts [2]    : 9971, 39%
(mean count < 10)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

## Principal Component Analysis (PCA)

```
vsd <- vst(dds, blind=FALSE)
plotPCA(vsd, intgroup=c("dex"))
```

using ntop=500 top features by variance

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the DESeq2 package.
  Please report the issue to the authors.
```



```
# PCA with ggplot2
```

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
pcaData <- plotPCA(vsd, intgroup=c("dex"), returnData=TRUE)
```

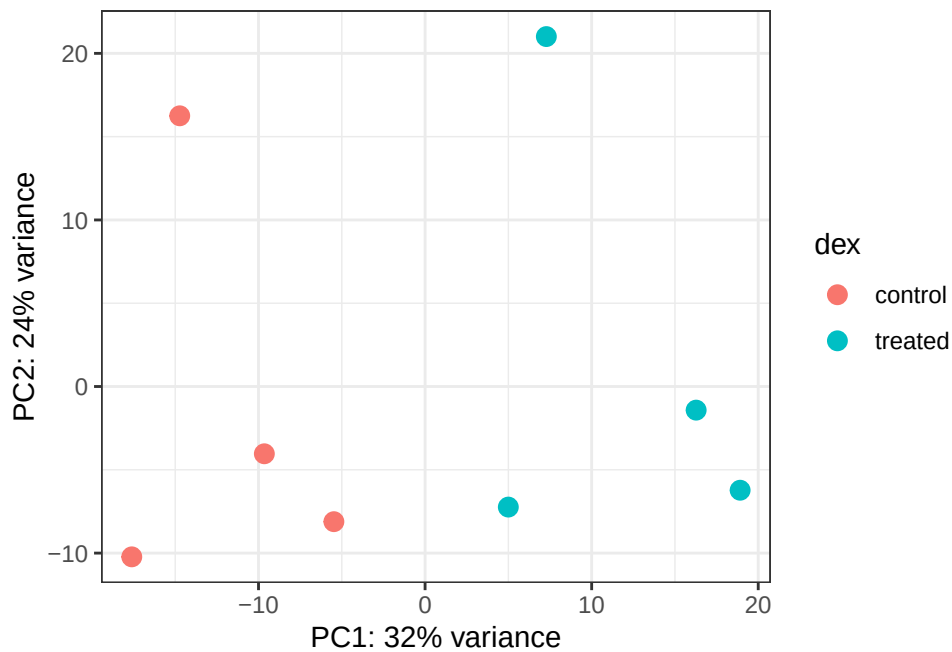
using ntop=500 top features by variance

```
head(pcaData)
```

|            | PC1        | PC2        | group   | dex     | name       |
|------------|------------|------------|---------|---------|------------|
| SRR1039508 | -17.607922 | -10.225252 | control | control | SRR1039508 |
| SRR1039509 | 4.996738   | -7.238117  | treated | treated | SRR1039509 |
| SRR1039512 | -5.474456  | -8.113993  | control | control | SRR1039512 |
| SRR1039513 | 18.912974  | -6.226041  | treated | treated | SRR1039513 |
| SRR1039516 | -14.729173 | 16.252000  | control | control | SRR1039516 |
| SRR1039517 | 7.279863   | 21.008034  | treated | treated | SRR1039517 |

```
percentVar <- round(100 * attr(pcaData, "percentVar"))

ggplot(pcaData) +
  aes(x=PC1, y=PC2, color=dex) +
  geom_point(size=3) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  coord_fixed() +
  theme_bw()
```



## DESeq analysis

```
res05 <- results(dds, alpha=0.05)
summary(res05)
```

```
out of 25258 with nonzero total read count
adjusted p-value < 0.05
LFC > 0 (up)      : 1236, 4.9%
LFC < 0 (down)    : 933, 3.7%
```

```

outliers [1]      : 142, 0.56%
low counts [2]    : 9033, 36%
(mean count < 6)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results

```

Using a more strict significant cutoff of  $\alpha=0.05$ , we still find many differentially expressed genes between treated and control samples.

## Adding annotation data

```

library(AnnotationDbi)
library(org.Hs.eg.db)

```

```

res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="SYMBOL",
                     multiVals="first")

```

'select()' returned 1:many mapping between keys and columns

```

res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="ENTREZID",
                     multiVals="first")

```

'select()' returned 1:many mapping between keys and columns

```

head(res)

```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 8 columns

|  | baseMean | log2FoldChange | lfcSE | stat | pvalue |
|--|----------|----------------|-------|------|--------|
|--|----------|----------------|-------|------|--------|

|                  | <numeric>  | <numeric>   | <numeric>   | <numeric> | <numeric> |
|------------------|------------|-------------|-------------|-----------|-----------|
| ENSG000000000003 | 747.194195 | -0.3507030  | 0.168246    | -2.084470 | 0.0371175 |
| ENSG000000000005 | 0.000000   | NA          | NA          | NA        | NA        |
| ENSG000000000419 | 520.134160 | 0.2061078   | 0.101059    | 2.039475  | 0.0414026 |
| ENSG000000000457 | 322.664844 | 0.0245269   | 0.145145    | 0.168982  | 0.8658106 |
| ENSG000000000460 | 87.682625  | -0.1471420  | 0.257007    | -0.572521 | 0.5669691 |
| ENSG000000000938 | 0.319167   | -1.7322890  | 3.493601    | -0.495846 | 0.6200029 |
|                  | padj       | symbol      | entrez      |           |           |
|                  | <numeric>  | <character> | <character> |           |           |
| ENSG000000000003 | 0.163035   | TSPAN6      | 7105        |           |           |
| ENSG000000000005 | NA         | TNMD        | 64102       |           |           |
| ENSG000000000419 | 0.176032   | DPM1        | 8813        |           |           |
| ENSG000000000457 | 0.961694   | SCYL3       | 57147       |           |           |
| ENSG000000000460 | 0.815849   | FIRRM       | 55732       |           |           |
| ENSG000000000938 | NA         | FGR         | 2268        |           |           |

Q11. Run the `mapIds()` function two more times to add the Entrez ID and UniProt accession and GENENAME as new columns called `res$entrez`, `res$uniprot` and `res$genename`.

```
res$uniprot <- mapIds(org.Hs.eg.db,
                      keys=row.names(res),
                      column="UNIPROT",
                      keytype="ENSEMBL",
                      multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$genename <- mapIds(org.Hs.eg.db,
                       keys=row.names(res),
                       column="GENENAME",
                       keytype="ENSEMBL",
                       multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res)
```

```
log2 fold change (MLE): dex treated vs control
Wald test p-value: dex treated vs control
DataFrame with 6 rows and 10 columns
```

|                   | baseMean   | log2FoldChange | lfcSE     | stat      | pvalue    |
|-------------------|------------|----------------|-----------|-----------|-----------|
|                   | <numeric>  | <numeric>      | <numeric> | <numeric> | <numeric> |
| ENSG000000000003  | 747.194195 | -0.3507030     | 0.168246  | -2.084470 | 0.0371175 |
| ENSG000000000005  | 0.000000   | NA             | NA        | NA        | NA        |
| ENSG0000000000419 | 520.134160 | 0.2061078      | 0.101059  | 2.039475  | 0.0414026 |
| ENSG0000000000457 | 322.664844 | 0.0245269      | 0.145145  | 0.168982  | 0.8658106 |
| ENSG0000000000460 | 87.682625  | -0.1471420     | 0.257007  | -0.572521 | 0.5669691 |
| ENSG0000000000938 | 0.319167   | -1.7322890     | 3.493601  | -0.495846 | 0.6200029 |

|                   | padj      | symbol      | entrez      | uniprot     |
|-------------------|-----------|-------------|-------------|-------------|
|                   | <numeric> | <character> | <character> | <character> |
| ENSG000000000003  | 0.163035  | TSPAN6      | 7105        | AOA087WYV6  |
| ENSG000000000005  | NA        | TNMD        | 64102       | Q9H2S6      |
| ENSG0000000000419 | 0.176032  | DPM1        | 8813        | H0Y368      |
| ENSG0000000000457 | 0.961694  | SCYL3       | 57147       | X6RHX1      |
| ENSG0000000000460 | 0.815849  | FIRRM       | 55732       | A6NFP1      |
| ENSG0000000000938 | NA        | FGR         | 2268        | B7Z6W7      |

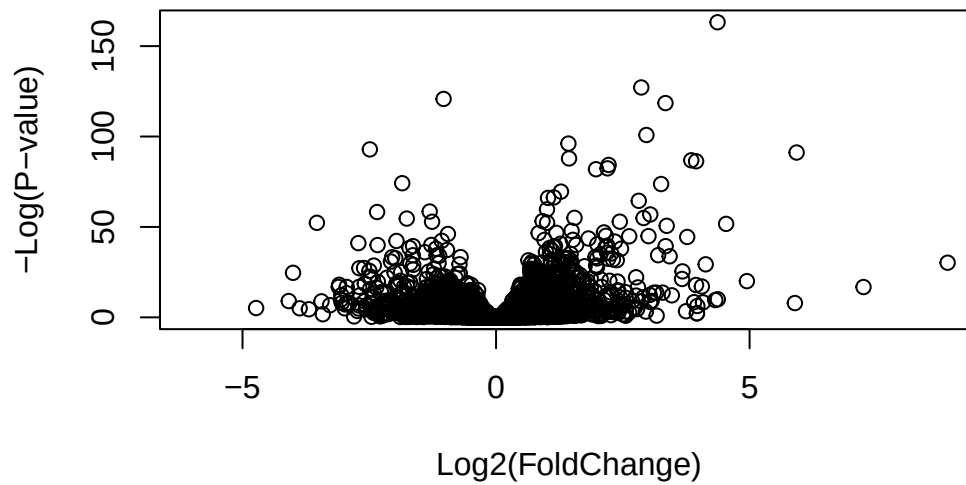
|                   | genename               |
|-------------------|------------------------|
|                   | <character>            |
| ENSG000000000003  | tetraspanin 6          |
| ENSG000000000005  | tenomodulin            |
| ENSG0000000000419 | dolichyl-phosphate m.. |
| ENSG0000000000457 | SCY1 like pseudokina.. |
| ENSG0000000000460 | FIGNL1 interacting r.. |
| ENSG0000000000938 | FGR proto-oncogene, .. |

Using `mapIds()` two more times with `column="UNIPROT"` and `column="GENENAME"` we can add UniProt accession numbers and full gene names to our results table.

## Data Visualization

```
plot(res$log2FoldChange, -log(res$padj),
     xlab="Log2(FoldChange)",
     ylab="-Log(P-value)")
```

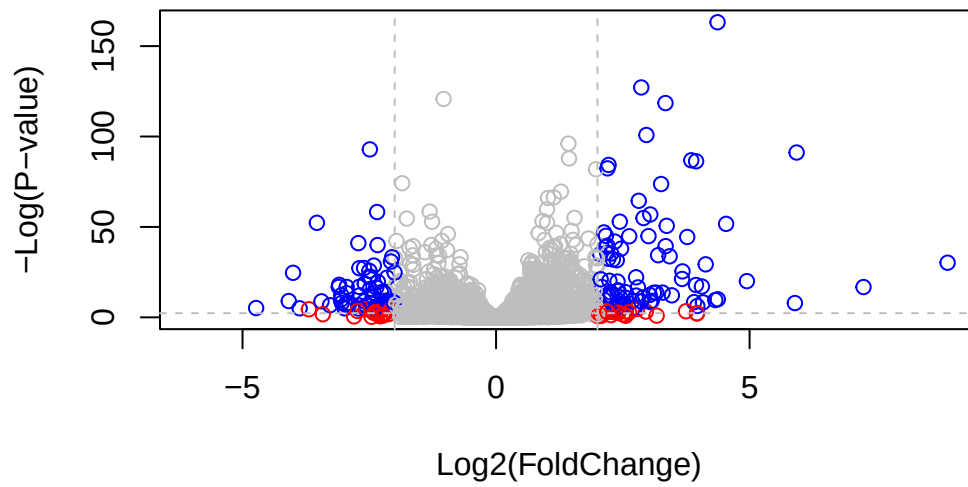




```
# Custom colored volcano plot
mycols <- rep("gray", nrow(res))
mycols[abs(res$log2FoldChange) > 2] <- "red"
inds <- (res$padj < 0.01) & (abs(res$log2FoldChange) > 2)
mycols[inds] <- "blue"

plot(res$log2FoldChange, -log(res$padj),
      col=mycols,
      ylab="-Log(P-value)",
      xlab="Log2(FoldChange)")

abline(v=c(-2,2), col="gray", lty=2)
abline(h=-log(0.1), col="gray", lty=2)
```



The volcano plot shows fold change vs statistical significance. Blue points are genes with both large fold change and high significance. Red points have large fold change but aren't statistically significant. Grey points don't meet either threshold.